Evaluation log

Test each piece in the order data flows through it. Record the date, the result and any fix.

#	Piece	How to test	Pass criteria	Date	Result	Notes / fix
1	Parsers	`npm test`	All tests green	2026-10-08	Pass	40/40 tests in the build environment
2	HFR client from Vercel	Admin → Check HFR portal (deployed app)	Rows returned, no 403/429			
3	Queue and runner	Pilot run spanning several ticks	No lost or duplicated tasks; survives a forced timeout	2026-10-08	Pass (local)	Validate run chained itself over HTTP across 3 ticks (after() + self-call); 0 failed tasks. Repeat on Vercel
4	Pilot scrape	Admin → Pilot (one region)	Facility count = portal's count for that region	2026-10-08	Pass (local)	Katavi: portal 200, stored 200, 195 assigned to a ward, 200 details fetched, 0 failed tasks
5	Full national load	Admin → Full national load	≥ 99 % of listed facilities stored with an internal ID; gaps explained			
6	Ward matching	Admin → Ward boundaries	≥ 95 % of HFR wards linked; rest linked by hand			2018 HDX file linked 41/54 Katavi wards: newer wards need manual links or a newer file
7	Ward check + sanity	Data quality → Outside its ward; hand-check 25	Flag correct in ≥ 90 %			Local Katavi run surfaced e.g. EREDITA (124533-1) placed 580 km from its ward. Added a 150 m edge tolerance after seeing points 2–112 m outside simplified 2018 polygons
8	OSM + Google matching	Hand-check 50 random graded facilities	Agree with grade in ≥ 90 %; Google under cap			
9	Geocoding missing points	Data quality → Missing coordinates	Every accepted point inside its ward; others flagged low			
10	Weekly diff	Two weekly runs ≥ 1 day apart	10 sampled changes all confirmed on the portal			Local weekly run 1 h after the pilot: 0 changes (expected), 50 stale details refreshed. Found and fixed a phase bug (numeric cast)
11	AI summary	Weekly reports → latest	Number check passes; nothing claimed that isn't in the data			Plain summary (no API key) generated and verified locally; Gemini path needs GEMINI_API_KEY
12	Email + cron	Wait for Monday's run (or trigger the cron route)	Email arrives; run starts within its hour			
13	Dashboard	Walk through every page	Pages load < 3 s; review decisions survive re-validation			

#	Piece	How to test	Pass criteria	Date	Result	Notes / fix
14	Climate catalogue	`npm test` (tests/climate.test.ts, tests/climate-api.test.ts)	69 unique services in the portal's exact category and name text, Kiswahili for each, all five tiers used	2026-10-21	Pass	151/151 tests; `tsc` and `eslint` clean. 8 services are marked `added` (not in the brainstorm) for the team to review
15	Migration 0010	`npm run db:migrate`, then `npx drizzle-kit generate`	Old climate rows cleared, `category_key` dropped, `service_translations` created; generate reports no changes	2026-10-21	Pass	Applied to the live Neon database. Deploys do not run migrations, so any other database needs `db:migrate` and then `classify-climate` by hand
16	Climate rebuild	`npx tsx scripts/classify-climate.ts`	Every catalogue service matches a portal row; every portal service has a Kiswahili name	2026-10-21	Pass	69 of 446 portal (category, service) pairs matched; 12,878 facilities; 119,272 facility-service links. First run left 10 pairs untranslated: a Windows text-encoding slip turned curly apostrophes and en-dashes into junk in the generated keys. Fixed and re-run: 0 untranslated
17	Kiswahili translations	Native-speaker health professional reviews the flagged names	Flagged names confirmed or corrected			342 service names and 44 categories drafted; 97 of 446 pairs flagged `needs_review` (mostly laboratory terms), none of the 69 climate services. Review outstanding
18	Climate dataset sheets	Build every sheet and the summary against the database	Sheets non-empty; Facility_Services rows = facility-service links; every exclusion has a reason	2026-10-21	Pass (local)	12,878 facilities in; 1,519 left out, all for no climate-sensitive service (1,002 have no services recorded). One facility has six services misfiled under Growth Monitoring that are not counted
19	Public API (live)	`curl /services`, `/facilities` (with `tier`, `service`, `lat+lng`, `lang=sw`) and an unknown code	69 services with CS codes, total 12,878, tier 5 = 1,617, Kiswahili names, 400 for an unknown code	2026-10-21	Pass	First check failed: pushes to `main` deploy as Preview only. Production follows `boundaries-arcgis-runner-fixes`, which was fast-forwarded to `92281e0`. The API cache keeps old answers up to 30 minutes, so test with a `?cb=` parameter
20	Dashboard export (Excel and CSV)	Export → Climate-sensitive facility dataset (deployed app)	Workbook and each CSV sheet have rows			Needs a signed-in session, so not tested from here. Run locally the same code builds 119,272 facility-service rows
21	API docs	Run each documented example against the live API	Every example and count matches	2026-10-21	Pass	Guide checked against production; Python example run live

#	Piece	How to test	Pass criteria	Date	Result	Notes / fix
22	Village centre grade	`npm test` (tests/geo.test.ts); migration 0011	A village centre is medium; ward and council centres stay low	2026-10-21	Pass	813 stored village centres regraded from low to medium by the migration
23	Village found by council	`npx tsx scripts/backfill-village-centres.ts`	Stand-ins with a village name move to that village's centre, no geocoder calls	2026-10-21	Pass	249 of 414 moved; 162 had no single matching village; 3 skipped (centre outside the council). Grades now: low 5,299 (was 6,361), medium 4,107 (was 3,039), village centres 1,062
24	MapTiler in use	Key test through the app code: search, reverse lookup, satellite picture	All three return data	2026-10-21	Pass (after 3 fixes)	Before the key was added no MapTiler call had ever been made. With the key: 403 until server calls sent the site's origin (the key is restricted to the Vercel site); reverse lookup returned 400 because the app sent `limit` (latent bug, fixed); the Static Maps API is not on this plan, so the picture is cut from satellite tiles (sharp). Search "Muhimbili", reverse names and a 512 px picture all work
25	Satellite check (plumbing)	Unit tests; local server: `/api/satellite`, facility page card, Issue filter; SQL run in a rolled-back transaction	Picture served, card shown, filter loads, statements valid	2026-10-21	Pass (local)	166/166 tests. 7,962 medium/low real points are candidates. Out-of-Tanzania points are rejected (400)
26	Satellite check (AI verdict)	Admin → Check points on satellite imagery, then eyeball 25 verdicts	Verdicts match what a person sees in at least 90 %	2026-10-21	Partly (spot test)	New Gemini key works (the old key's credit was out). Spot test: known facility → building; open grassland → none; ocean (black tile) → unclear. Models answer 503 (busy) at times, so the check retries 3 times (2, 6, 15 s) then stops until the next run; calls take 8 to 38 s. `gemini-3.1-flash-lite` says "none" on empty land where `gemini-3.8-flash` says "unclear". The 25-point eyeball check is still to do

Notes: climate-sensitive services (2026-10-21)

- **Decision.** The climate data is the normal registry data filtered, not a renamed copy. Each climate service is a portal service exactly as filed (category and name unchanged, wrong categories and spelling kept and noted); only Kiswahili, the five tiers and a short description are added. Codes `CS001` to `CS069` are catalogue ids because the portal gives services none.

- **Matching** is by the portal's exact (category, service name) pair, so every climate row traces to a `facility_services` row. The same service filed under several categories has one code per category.
- **Breaking change.** The public API now returns `CS` codes and portal names. Anything built on the old codes (for example `MAL-MRDT`) must be updated.
- **Correction.** Growth monitoring does exist in the portal (maternity and nursing homes, under 40 facilities) and is included in tier 4.
- **Open items.** Native-speaker review of the Kiswahili; whether toilets, wash slab, incinerator and waste services join tier 4; how the app treats the 44 % of facilities with low-confidence locations; the one facility with misfiled services; whether to hide phone numbers in the open API.
- **Reference:** `docs/Climate-Dataset-Summary-Report` and `docs/Climate-API-Guide` (Markdown, Word and PDF, kept local because `docs/` is gitignored).

Notes: point quality and the satellite check (2026-10-21)

- **Village centres are medium.** A facility with no usable point is placed on the centre of its HFR village and graded medium (right village, not on the building). The village is matched by ward, or by name inside the council when the ward name is out of date. Ward and council centres stay low.
- **Satellite check.** Real medium/low points get a satellite picture (MapTiler key if set, else Google Static Maps) and Gemini says whether a building is at the centre. A building lifts the grade one step; no building or unclear never lowers it and goes to the review queue (Data quality → Issue). Stand-in centres are not checked: a building at a village centre proves nothing.
- **Open items.** Add a MapTiler key (separate from Google) if MapTiler geocoding and imagery are wanted; top up Gemini credit; enable Maps Static API billing/limits on the Google key and set `GOOGLE_STATIC_MONTHLY_CAP`; check Google's terms for sending Static Maps pictures to a model; the 382 ward/council stand-ins with no village name or no match.